

1. (a) (i) For $x = 1$, $f_n(1) = f(1) = 0$. So $f_n(1) \rightarrow f(1)$ as $n \rightarrow \infty$.
 For $x = 0$, $f(0) = 0$ and $f_n(0) = n$, $\forall n$. So $f_n(0) \nrightarrow f(0)$ as $n \rightarrow \infty$.
 For $x \in (0, 1)$, then $\exists n_0$ s.t. $\frac{1}{n_0} < x$ and $\forall n \geq n_0$, $f_n(x) = 0 = f(x)$. So $f_n(x) \rightarrow f(x)$, $x \in (0, 1)$ as $n \rightarrow \infty$.
 So $\{x : f_n(x) \nrightarrow f(x)\} = \{0\}$ and $m(\{0\}) = 0$. Hence $f_n \rightarrow f$ a.e. (μ).
- (ii) Since $\int |f_n| I_{\{|f_n| > T\}} dm = nm([0, \frac{1}{n})) = 1$, $\forall n > T$, then $\sup_{n \geq 1} \int |f_n| I_{\{|f_n| > T\}} dm \geq 1$.
 1. So $\lim_{T \rightarrow \infty} \sup_{n \geq 1} \int |f_n| I_{\{|f_n| > T\}} dm \neq 0$. Hence f_n is not UI.
- (iii) $\int |f_n - f| dm = \int n I_{[0, \frac{1}{n})} dm = n \cdot \frac{1}{n} = 1 \nrightarrow 0$. So $f_n \nrightarrow f$ in L^1 .
- (b) If $\{f_n^p\}$ is UI, i.e. $\lim_{T \rightarrow \infty} \sup_{n \geq 1} \int |f_n| I_{\{|f_n| > T\}} dm = 0$, then $f_n \rightarrow f$ a.e. (m) would imply $f_n \rightarrow f$ in L^p (HW problem: 2.36(b)) since $m(\Omega) = m([0, 1]) = 1 < \infty$. There are other correct answers too, for example, along the lines of DCT.
- (c) Note that in this case $r = p/q > 1$ and hence $\phi(x) = x^r$, $x \in [0, 1]$ is a convex function. Using Jensen's, we have

$$\left(\int |f_n - f|^q d\mu \right)^{\frac{p}{q}} \leq \left(\int |f_n - f|^{q \frac{p}{q}} d\mu \right) = \left(\int |f_n - f|^p d\mu \right).$$

or in other words, $\|f_n - f\|_q \leq \|f_n - f\|_p \rightarrow 0$, as $n \rightarrow \infty$.

2. (a) Using Markov's inequality, for each $t > 0$,

$$P(X > a) = P(e^{tX} > e^{ta}) \leq e^{-ta} M(t),$$

and hence the result follows.

- (b) Apply Cauchy-Schwarz to $X I_{\{X > 0\}}$ to obtain

$$E[X I_{\{X > 0\}}] \leq [E(X^2)]^{1/2} [P(X > 0)]^{1/2}$$

using the fact that $I_{\{X > 0\}}^2 = I_{\{X > 0\}}$ and $E[I_{\{X > 0\}}] = P(X > 0)$. Also, since X is nonnegative, $E(X) = E(X I_{\{X=0\}}) + E(X I_{\{X>0\}}) = E(X I_{\{X>0\}})$. Hence, squaring both sides above, we get $[E(X)]^2 \leq E(X^2) P(X > 0)$ and the result follows.

3. (a) Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and f_n, f, g be measurable functions. If $|f_n| \leq g$, a.e. (μ), $\forall n \geq 1$, $\int g d\mu < \infty$ and $f_n \rightarrow f$ a.e. (μ), then $f \in L^1(\Omega, \mathcal{F}, \mu)$, $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$ and $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.
- (b) We need to prove the following modified DCT:
 Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and f_n, f, g be measurable functions. If $|f_n| \leq g$, a.e. (μ), $\forall n \geq 1$, $\int g d\mu < \infty$ and $f_n \rightarrow f$ in measure, then $f \in L^1(\Omega, \mathcal{F}, \mu)$, $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$ and $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.

Proof: First we state two facts in real analysis:

Fact 1: A sequence $\{x_n\}$ converges to x iff for any subsequence $\{x_{n_k}\}$ of $\{x_n\}$, there is a further subsequence $\{x_{n_{k_l}}\}$ of $\{x_{n_k}\}$ such that $x_{n_{k_l}} \rightarrow x$.

Fact 2 (Theorem 2.5.2): $f_n \rightarrow f$ in measure $\Rightarrow$ there exists a subsequence $\{f_{n_k}\}$ such that $f_{n_k} \rightarrow f$ a.e. (μ).

Then we prove $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$.

Denote $a_n = \int f_n d\mu$, $a = \int f d\mu$ and $b_n = \int |f_n - f| d\mu$. To show $a_n \rightarrow a$ and $b_n \rightarrow 0$, we need to use Fact 1. Choose any subsequence $\{a_{n_k}\}$ of $\{a_n\}$, and for this subsequence, choose the corresponding subsequence functions $\{f_{n_k}\}$ of $\{f_n\}$. Since $f_n \rightarrow f$ in measure, then $\lim_{n \rightarrow \infty} \mu(|f_n - f| > \varepsilon) = 0$, $\forall \varepsilon > 0$, thus $\lim_{k \rightarrow \infty} \mu(|f_{n_k} - f| > \varepsilon) = 0$, $\forall \varepsilon > 0$, i.e. $f_{n_k} \rightarrow f$ in measure. By Fact 2, we can get a sub-subsequence $f_{n_{k_l}} \rightarrow f$ a.e. (μ). For this sub-subsequence $\{f_{n_{k_l}}\}$, conditions of DCT are true. Hence from DCT, we have $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\lim_{l \rightarrow \infty} \int f_{n_{k_l}} d\mu = \int f d\mu$, i.e. $a_{n_{k_l}} \rightarrow a$. By Fact 1, we get $a_n \rightarrow a$, i.e. $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$. Similarly, we have $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.

4. (a) $\mu_3(\emptyset) = \mu_1(\emptyset) + \mu_2(\emptyset) = 0$;
 $\mu_3(A) = \mu_1(A) + \mu_2(A) \geq 0 \Rightarrow \mu_3(A) \in [0, \infty]$, $\forall A \in \mathcal{F}$;
 If $\{A_i\}$ is a disjoint collection of sets in $\mathcal{F}$, then $\mu_3(\cup_i A_i) = \mu_1(\cup_i A_i) + \mu_2(\cup_i A_i) = \sum_i \mu_1(A_i) + \sum_i \mu_2(A_i) = \sum_i (\mu_1(A_i) + \mu_2(A_i)) = \sum_i \mu_3(A_i)$.
 So μ_3 is a measure.
- (b) $\nu(A) = 0 \Rightarrow \mu_1(A) = \mu_2(A) = 0 \Rightarrow \mu_3(A) = \mu_1(A) + \mu_2(A) = 0$. So $\mu_3 \ll \nu$.
- (c) Claim: $\frac{d\mu_3}{d\nu} = \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$ a.e. (ν). *Proof:* Since $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\exists f_1, f_2 \geq 0$ s.t. $\mu_1(A) = \int_A f_1 d\nu$, $\mu_2(A) = \int_A f_2 d\nu$. By definition, $\mu_3(A) = \mu_1(A) + \mu_2(A) = \int_A f_1 d\nu + \int_A f_2 d\nu = \int_A (f_1 + f_2) d\nu$ by linearity of integrals. Hence by uniqueness of RN derivatives, $\frac{d\mu_3}{d\nu} = f_1 + f_2 = \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$ a.e. (ν).
- (d) $\mu \perp \nu \Rightarrow \exists B \in \mathcal{F}$ s.t. $\nu(B) = \mu(B^c) = 0$. Since $\mu \ll \nu$ and $\nu(B) = 0$, then $\mu(B) = 0$. So $\mu(B) = \mu(B^c) = 0 \Rightarrow \mu(\Omega) = 0$. Thus $\forall A \in \mathcal{F}$, $0 \leq \mu(A) \leq \mu(\Omega) = 0 \Rightarrow \mu(A) \equiv 0$, $\forall A \in \mathcal{F}$, i.e. $\mu \equiv 0$.
- (e) Let $B = \mathbb{N} \cup \{0\} = \mathbb{N}_0$, then $\mu_1(B) = 0 = \nu(B^c)$, i.e. $\mu_1 \perp \nu$. Choose $\mu_s = \mu_1$, $\mu_a = \mu_2$ and $\mu_3 = \mu_a + \mu_s$. To show $\mu_a \ll \nu$ and compute $\frac{d\mu_a}{d\nu}$.
 Let $\mu_c(A) = |A \cap \mathbb{N}_0|$, $\forall A$ which defines a counting measure on $\mathbb{N}_0$. From table 4.6.1, $\nu(A) = 0 \Rightarrow A \cap \mathbb{N}_0 = \emptyset \Rightarrow \mu_a(A) = 0$, i.e. $\mu_a \ll \nu$. So μ_a is the absolute continuous part of μ_3 w.r.t. ν . Then $\mu_2(A) = \sum_{x \in \mathbb{N}_0} I_A(x)(0.7)^x 0.3 = \int p_G(x) d\mu_c(x)$ where $p_G(x) = I_{\mathbb{N}_0}(x)(0.7)^x 0.3$. Similarly $\nu(A) = \int p_P(x) d\mu_c(x)$ where $p_P(x) = e^{-8} 8^x (x!)^{-1} I_{\mathbb{N}_0}(x)$.

Since $p_P(x) > 0$ a.e. (μ_c) , then

$$\begin{aligned}
\frac{d\mu_a}{d\nu}(x) &= \frac{d\mu_2}{d\mu_c} \left(\frac{d\nu}{d\mu_c} \right)^{-1} \\
&= \left(I_{\mathbb{N}_0}(x) (0.7)^x 0.3 \right) \cdot \left(I_{\mathbb{N}_0}(x) e^{-8} 8^x (x!)^{-1} \right)^{-1} \\
&= \frac{(0.7)^x 0.3 e^8 x!}{8^x} I_{\mathbb{N}_0}(x), \text{ a.e. } (\nu).
\end{aligned}$$